Final Project Deliverables

AI churn prediction for Ravenstack

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AI Consulting Bootcamp · Final Project

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1. Use case definition

1.1 Business problem statement

Ravenstack loses approximately $605,000 in ARR each year to customer churn. The customer success team (1 to 3 people) manages 390 active accounts with no early warning system. They decide who to call based on recency and gut feel.

The data shows 53% of all churn happens in the first 90 days after signup. By the time the CS team notices an account is in trouble, the account has usually already decided to cancel. DevTools accounts churn at 31%, twice the rate of the lowest-churn segment. There is no mechanism to surface this risk before the cancellation arrives.

The problem is not that the team lacks effort. The data to predict churn already exists in Ravenstack's own systems. The gap is that no one has built a process to turn that data into a weekly ranked list of which accounts to call.

1.2 Company profile

Field	Detail
Company name	Ravenstack
Industry	B2B SaaS (software subscriptions)
Company size	Small-to-mid-market, 200 employees
Annual recurring revenue	Approximately $60M
Active accounts	390 (out of 500 total; 110 have churned over the 2023-2024 period)
Annual churn rate	12.1% (measured from the company's own data)
Customer success team	1 to 3 people, no dedicated data or analytics function
Technology stack	Subscription billing system, usage logging, Zendesk for support, Slack for internal comms
Current retention method	Ad hoc outreach triggered by cancellation notices or long periods of silence

1.3 Proposed AI solution

A weekly churn risk scoring and alert system with 3 components:

- **Scoring (n8n Code node):** scores all 390 active accounts weekly using a rule-based model (tenure under 90 days, DevTools industry, usage drop, billing frequency). The scoring logic runs inside an n8n Code node, so there is no separate server to host or keep running. It returns a ranked list of the top 20 at-risk accounts with their risk signals. Designed to be replaced by a machine learning model in Phase 3 if the model outperforms the rules.

- **Claude LLM (Anthropic API):** receives the top 20 account list and writes one plain-English sentence per account explaining why it was flagged and what the CS manager should do. One API call per weekly run.
- **n8n automation workflow:** runs every Monday at 8am, reads the account data from URLs, scores every account in a Code node, passes the results to Claude, formats the output as Slack Block Kit blocks, and posts to the #customer-success channel. The whole pipeline lives in one workflow with no external dependencies. It is exported as a .json file and importable into any n8n instance.

AI system type: Recommendation (decision-support). The system ranks accounts and explains the ranking. The customer success team makes all contact decisions.

1.4 Key stakeholders

Stakeholder	Role in this system
Cleo (CEO)	Approves pilot budget and timeline. Receives monthly reports on alert quality and churn rate changes. The primary audience for the business case.
CS team lead	Primary user. Reviews the weekly digest every Monday morning. Rates each alert as relevant or irrelevant. Assigns accounts to team members for outreach.
CS team members	Act on the alerts. Contact flagged customers. Leave notes in Slack thread.
Engineering (0.5 FTE)	Maintains the scoring API and n8n workflow. Runs monthly model retraining. Monitors drift detection alerts.
Customers (data subjects)	Their behavioral data is processed by the scoring model. They are not contacted automatically. All contact decisions rest with the CS team.

1.5 Success criteria

Two measurable outcomes define a successful pilot:

- **Precision threshold:** The CS team rates at least 50% of flagged accounts as 'relevant' (meaning the team would have prioritised that account for outreach based on the information provided) in blind testing at week 4 of the pilot. This is measured by tracking Slack reactions on each alert.
- **Retention lift:** Flagged accounts that received CS outreach show a monthly churn rate at least 15% lower than the control group (accounts flagged but not contacted) at week 8. This requires the CS team to deliberately leave some flagged accounts uncontacted for measurement purposes.

If precision falls below 50% after 4 weeks, the pilot stops and the team investigates the scoring model before restarting. If retention lift is below 15% after 8 weeks, the team assesses whether the issue is the model or the outreach approach.

1.6 Out-of-scope boundaries

This system explicitly does not:

- Send automated emails or messages to customers. All contact decisions rest with the CS team.
- Score trial accounts or free-tier users. Only paid, active accounts are included.
- Perform real-time scoring. The system runs once per week. There is no event-triggered scoring.
- Predict expansion revenue, upsell potential, or product fit. That is a separate use case planned for the acquisition roadmap.
- Make any employment-related decisions about Ravenstack's own staff.
- Run on customer-facing surfaces. Customers do not interact with the AI.

2. No-code / low-code proof of concept

2.1 Tools used and why

Tool	Role and reason for selection
n8n (self-hosted)	Workflow automation. Handles the weekly schedule, fetching the data from URLs, the scoring, the conditional logic, the Claude call, and the Slack post. Free on self-hosted. The entire workflow is 11 nodes and imports from a single JSON file.
n8n Code node (JavaScript)	Scoring. The model logic runs in a Code node inside n8n, so there is no separate API server to host or keep running. It parses the CSV files, joins them, and scores every account. No ML framework is needed for the rule-based Phase 1 model.
Anthropic API (Claude)	LLM explanation generation. One API call per weekly run (all 20 accounts in one prompt). Generates one plain-English sentence per account. Estimated cost: $0.10 to $0.30 per run.
Slack Incoming Webhook	Alert delivery. No OAuth required. Creates a webhook URL in the Slack API settings; n8n posts to it directly. The CS team sees the digest in their existing Slack workspace.

2.2 What the POC does (step by step)

Step 1: The n8n Schedule Trigger fires at 08:00 UTC every Monday.

Step 2: three HTTP nodes fetch the Ravenstack CSV files (accounts, subscriptions, usage) from their public URLs. A Code node then parses them, joins them, scores all 390 active accounts with the rule-based model, and outputs the top 20 as a JSON array with account name, industry, plan, MRR, risk score, and risk signals. There is no external API call; the scoring happens inside n8n.

Step 3: An IF node checks that the response contains at least one account. If it does not, the workflow takes the false branch (Step 8).

Step 4: A Code node formats the 20 accounts into a structured prompt for Claude. The prompt asks Claude to write one sentence per account, constrained to 20 words, focused on what the CS manager should do.

Step 5: n8n sends a POST request to api.anthropic.com/v1/messages with the prompt. Claude returns a numbered list of 20 sentences.

Step 6: A Code node parses Claude's response and builds Slack Block Kit blocks. Each account becomes a single section block showing the company details, risk score, and Claude's one-sentence explanation. Keeping one block per account stays within Slack's 50-block-per-message limit.

Step 7: n8n posts the formatted blocks to the Slack Incoming Webhook URL. The digest appears in #customer-success within seconds.

Step 8 (false branch): If no accounts were returned, n8n posts a short 'all clear' text message to Slack so the team knows the workflow ran.

2.3 What AI capability is demonstrated

Claude receives structured data (account name, behavioral signals) and converts it into plain-English action advice for a non-technical reader. Example:

```
Input: Company_321, DevTools, Basic, $171/mo, 6 days old, usage down 100%,
last login 36 days ago
Output: 'Six-day-old DevTools account with no activity in 5 weeks — call
before the 30-day mark or this account likely churns.'
```

The LLM step does not predict churn. The scoring model does that. Claude translates the model's output from data into language the CS manager can act on in under 30 seconds.

2.4 Known limitations vs a production system

- **No A/B control group tracking:** The POC posts alerts but does not track whether the CS team contacted the account or record the outcome. A production system would log this in the CRM.
- **Rule-based model only:** The Phase 1 scoring model applies fixed rules. It will miss patterns a trained classifier would detect. The Phase 3 plan addresses this.
- **No CRM integration:** Account assignments (which CS rep owns which account) are not connected. The digest does not route to the correct rep automatically.
- **No feedback loop:** CS reps can react with emojis in Slack, but those reactions are not automatically captured to improve the model. A production system would read Slack reactions via the Slack API and write them to a feedback table.
- **LLM explanations are bounded by scoring API inputs:** If the scoring Code node misses a signal, Claude cannot surface it. The LLM explains the signals it receives; it does not independently analyse account data.

3. ROI and risk assessment

3.1 Cost estimates

Upfront costs

Item	Cost	Notes
Developer time: Phase 1 data audit (2 weeks at 0.5 FTE)	$1,600	At $80/hr European mid-market rate
Developer time: Phase 2 pilot model (4 weeks at 0.5 FTE)	$6,400	Same rate
Developer time: Phase 3 integration (4 weeks at 0.5 FTE)	$1,800	Lighter load in Phase 3
Tooling setup (n8n self-hosted VM, initial config)	$200	One-time
Total upfront	**$10,000**	Midpoint of the $8,000-$12,000 range

Ongoing monthly costs

Item	Monthly cost	Notes
Cloud compute (batch scoring jobs)	$55	Midpoint of $30-$80; small VM or serverless
Anthropic API (LLM, 52 runs/year)	$125	Midpoint of $50-$200; ~$0.10-$0.30 per run
Tooling (n8n cloud, Power BI Pro)	$65	Midpoint of $30-$100
CS team review time (2 hrs/week at $50/hr)	$400-$800	Staff cost already budgeted; shown for transparency
Monthly retraining and drift monitoring	$1,150	Midpoint of $800-$1,500; includes compute and engineer time
Total monthly (excluding CS staff cost)	**$1,395**	Midpoint of $1,310-$2,680 excluding staff

3.2 Business value estimate

Ravenstack's annual ARR is approximately $60M with 390 active accounts. The annual churn rate measured from the data is 12.1%.

Metric	Value
ARR at risk annually (12.1% × $60M)	$7.26m
Accounts churning per year (12.1% × 390)	~47 accounts
Average account ARR ($60M / 390)	$153,846 per account per year
If CS outreach saves 15% of at-risk accounts (7 accounts)	$1.09m ARR retained per year

The 15% retention improvement is a conservative assumption. Published B2B SaaS case studies (HubSpot, ProfitWell, ChartMogul) show 5% to 25% improvement when usage-based alerts are paired with CS outreach at SMB scale. Using the lower end of that range protects the ROI calculation from optimism.

3.3 ROI calculations

12-month ROI

Assumptions: implementation takes 3 months (Phases 1-3). Benefit starts accruing from month 4.

Item	Value
Upfront cost	$10,000
Monthly ongoing cost (months 4-12, 9 months)	$12,555 ($1,395 × 9)
Total year 1 cost	$22,555
Year 1 benefit (benefit starts month 4: $90,750/mo × 9 months)	$816,750
Net benefit (year 1)	$794,195
ROI at 12 months = (Net benefit / Total cost) × 100	3,521%

36-month ROI

Item	Value
Upfront cost (one-time)	$10,000
Monthly ongoing cost (months 4-36, 33 months)	$46,035
Total 36-month cost	$56,035
36-month benefit ($90,000/year × 3 years)	$3.27m
Net benefit (36 months)	$3.213m
ROI at 36 months = (Net benefit / Total cost) × 100	5,735%

3.4 Assumptions table

Assumption	Value used	Justification
Company ARR	$60M	Consistent with $40M-$80M range for 200 person B2B SaaS
Annual churn rate	12.1% ($7.26m)	Measured directly from Ravenstack's own 2023-2024 data
CS outreach save rate	15% ($1.09m)	Conservative end of the 5%-25% range reported in SaaS case studies
Average account ARR	$153,846	Total ARR divided by 390 active accounts
Monthly ongoing cost	$1,395	Midpoint of the estimated range, excluding staff cost already budgeted
Upfront cost	$10,000	Midpoint of the $8,000-$12,000 developer cost estimate
Implementation timeline	3 months	Based on the 10-14 week estimate across Phases 1-3
Benefit start	Month 4	After full deployment at the end of Phase 3
Developer rate	$80/hr	European mid-market rate; US market rate would raise upfront to $12,000-$15,000
Benefit applies to full 12 months from year 2 onward	Yes	Assumes the system stays live and CS team continues using it

3.5 Break-even point

Monthly net benefit once live: $90,750 revenue protected minus $1,395 ongoing cost = $89,355 per month net.

Time to recover the $10,000 upfront investment from the net monthly surplus: $10,000 / $89,355 = less than 1 month after go-live.

The system goes live at month 3 (end of Phase 3). Break-even is approximately month 3 -4 from project start.

3.6 Risk assessment matrix

Risk	L	I	Score	Mitigation
Weak signal in data: the scoring model's risk signals may not correlate with actual churn in Ravenstack's data.	3	3	9	Phase 1 go/no-go decision. Build a rule-based health score as a baseline. Ship rules and stop ML if the model fails to beat them.
Model drift: a new product feature or pricing change	4	3	12	Monthly retraining. Weekly feature drift detection: flag any input feature that shifts more than 20% from training distribution.

shifts the behavioral patterns the model was trained on.				
False positive overload: the model flags too many low-risk accounts and the CS team stops trusting the digest.	3	2	6	Success criterion #1 (50% precision at week 4) stops this in the pilot. Precision threshold set before Phase 3 go-live.
GDPR non-compliance: account behavioral data sent to the Anthropic API (US-based) without adequate safeguards.	2	4	8	Review and sign Anthropic's Business API Data Processing Agreement. Use account IDs rather than company names in prompts if the DPA review flags concerns. No individual personal data in prompts.
Biased recommendations: the scoring model weights DevTools accounts at +25 points, potentially over-flagging that segment.	2	3	6	Control group in the pilot. Compare actual churn rates by industry against model predictions after 8 weeks. Recalibrate weights if DevTools flagging does not predict actual churn.
Low CS team adoption: the team ignores the alerts or finds the digest too long to review weekly.	3	4	12	CS team involved in rating alerts from week 1. Digest designed to take 5 minutes to review. Track Slack reaction rate as a usage metric. Stop the workflow if reaction rate falls below 50% for 2 consecutive weeks.
LLM hallucinated alert text: Claude generates plausible but factually incorrect explanations.	3	3	9	Prompt tightly constrained to signals provided by the scoring API. Raw signals shown below each LLM line so the CS manager can cross-check. CS team trained to treat explanations as suggestions.
API cost overrun: n8n misconfiguration triggers repeated scoring runs and unexpected API charges.	2	1	2	Workflow runs on a controlled cron schedule. Monthly API spend alert set at $300. Budget cap configured in the Anthropic account dashboard.

L = Likelihood (1 = very unlikely, 5 = very likely). I = Impact (1 = negligible, 5 = severe). Score = L × I.

Scores of 10 or above warrant a formal mitigation plan and a named owner. Scores below 6 are logged and reviewed quarterly.

4. EU AI Act compliance documentation

4.1 Risk classification

Classification: Limited Risk, under Article 52 of the EU AI Act (Regulation 2024/1689, entered into force August 2024).

4.2 Classification reasoning

The EU AI Act defines four risk levels: Unacceptable, High, Limited, and Minimal.

Unacceptable risk: does not apply

The system does not use subliminal manipulation, social scoring, real-time biometric surveillance, or any of the prohibited practices in Article 5.

High risk: does not apply

Annex III of the Act lists high-risk AI systems. The relevant categories to check:

- **Employment (Annex III, 4b):** Covers AI used to make decisions affecting workers (hiring, promotion, monitoring). This system scores B2B customer accounts, not employees. It does not apply.
- **Essential services (Annex III, 5):** Covers creditworthiness, health and life insurance. Customer retention scoring for a SaaS product does not fall within this category.
- **Law enforcement, migration, education, critical infrastructure:** None apply.

The system is a B2B internal tool used by trained staff to support customer retention decisions. The CS team makes all contact decisions. The model makes no binding decisions.

Limited risk: applies

The system includes a large language model (Claude) that generates text visible to the CS team. Under Article 52(1), users of AI systems that interact with or are visible to humans must be informed that AI generated the content.

The system already meets this obligation: every Slack digest displays 'Model: rule-based v1' in the header, and the LLM-generated explanations are visibly labeled as such.

Why not Minimal risk

Minimal risk carries no obligations. Given the LLM component and its direct influence on which customers the CS team contacts (a commercial decision affecting third parties), Limited Risk is the more defensible and responsible classification.

4.3 Mandatory requirements summary (Limited Risk)

Requirement	How the system addresses it
Article 52(1): disclose that the digest is AI-generated to users	Every Slack digest header reads 'Model: rule-based v1'. A permanent note will be added to the #customer-success channel description before Phase 3 goes live.
Article 52(2): emotion recognition disclosure	Not applicable. The system does not use emotion recognition.
Article 52(3): deepfake disclosure	Not applicable. The system does not generate synthetic images or video.
Human oversight	All contact decisions rest with the CS team. The workflow posts recommendations; no automated contact with customers occurs. This is documented in the operating procedure given to the CS team.

4.4 Conformity assessment summary

What the system does

The Ravenstack Churn Risk Digest is a decision-support AI system. It scores 390 B2B customer accounts weekly for churn risk using a rule-based algorithm, generates plain-English explanations of the risk signals using a large language model, and delivers a ranked alert list to the customer success team via Slack every Monday morning.

Risk class and basis

Limited Risk under Article 52, EU AI Act 2024/1689. The system is an internal B2B tool with LLM-generated content visible to trained staff. It does not meet the Annex III high-risk criteria. The LLM component triggers Article 52(1) transparency obligations.

Applicable obligations and how the design addresses them

Article 52(1) is addressed by the 'Model: rule-based v1' disclosure in every digest header. No automated contact with customers occurs, so no further disclosure to data subjects is required at this stage.

Gaps to resolve before deployment

- Written policy for the CS team explaining what the model does and does not do. To be created before Phase 3 go-live.
- Slack channel description updated to note that alerts are AI-generated.
- Anthropic API Data Processing Agreement reviewed, signed, and stored on file.
- Process for the CS team to report incorrect or harmful alerts. Currently no formal feedback channel exists.

4.5 Technical documentation outline

The following is a table of contents for the full technical documentation package that would be required if the system were reclassified as High Risk, or if it is productized and offered to third-party customers. The documents do not need to be written now.

1. System description

- 1.1 Name, version, and intended use
- 1.2 Component list (scoring Code node, n8n workflow, LLM API, Slack)
- 1.3 Intended users and use environment

2. Risk management file

- 2.1 Risk identification process
- 2.2 Risk scores and residual risk after mitigation

3. Training and validation data

- 3.1 Data sources (Ravenstack billing, usage logs, support tickets)
- 3.2 Data quality assessment
- 3.3 Known gaps (weak usage drop signal, synthetic dataset limitations)

4. Model design

- 4.1 Scoring algorithm design and feature selection
- 4.2 Validation methodology (time-based train/test split)
- 4.3 Phase 2 ML model design (if rules are replaced)

5. Performance metrics

- 5.1 Pilot precision and recall results
- 5.2 Drift detection thresholds and monitoring plan

6. Human oversight documentation

- 6.1 CS team decision authority and operating procedure
- 6.2 Alert response protocol
- 6.3 Feedback and correction mechanism

7. Incident management

- 7.1 Process for reporting incorrect or harmful alerts
- 7.2 Escalation path and logging

5. GDPR documentation

5.1 Data flow map

Source	Data sent	Processing step	Destination
Ravenstack billing system	Account ID, company name, industry, country, plan tier, MRR, billing frequency	Risk scoring (tenure and billing signals)	Scoring API (internal)
Ravenstack product database	Feature usage counts, usage dates, error counts	Risk scoring (usage drop signal)	Scoring API (internal)
Ravenstack support system	Ticket counts, satisfaction scores	Supplementary signal only (not in current Phase 1 model)	Scoring API (internal)
Scoring API	Top 20 account records with risk signals (name, industry, plan, MRR, signals)	LLM explanation generation	Anthropic API (USA)
Anthropic API	Plain-English explanation text (20 sentences)	Slack digest formatting	n8n (internal)
n8n workflow	Formatted Slack Block Kit payload	Delivery to CS team	Slack (USA)

Personal data note: account names in the dataset follow a 'Company_NNN' pattern in the test data and are B2B company identifiers, not individual names. In production, if an account belongs to a sole trader whose company name is their personal name, that name is personal data under GDPR Article 4(1). The system must treat such records accordingly.

5.2 Processing activities register

Data type	Purpose	Legal basis	Retention	Third parties
Account identifiers (ID, name, industry, country)	Score accounts for churn risk	Legitimate interest (Art. 6(1)(f))	90-day rolling window in scoring API; logs deleted after each run	None
Subscription data (plan, MRR, billing frequency)	Calculate billing-related risk signals	Legitimate interest	Same as above	None

Usage logs (feature counts, dates)	Calculate usage-based risk signals	Legitimate interest	90-day rolling window	None
Formatted account data with risk signals	LLM explanation generation	Legitimate interest	Sent per run; not retained by Anthropic (per Business API DPA)	Anthropic (USA, SCCs)
Slack alert content (names, scores, explanations)	Deliver alerts to CS team	Legitimate interest	Subject to Slack's retention policy (90 days on free plan)	Slack Technologies (USA, SCCs)

Legitimate interest assessment (LIA)

Purpose: Prevent customer churn by giving the CS team early warning of at-risk accounts.

Necessity: Processing customer usage data is necessary to generate churn risk scores. No less invasive method achieves the same result.

Balancing test: Accounts are B2B entities. Individual users are not directly identified or contacted. Data is used only to help the account by triggering proactive CS support. Human review precedes all contact decisions.

5.3 Data Protection Impact Assessment (DPIA)

Highest-risk processing activity: sending account data to the Anthropic API.

Description of the processing

Once per week, the scoring Code node compiles a structured list of up to 20 account records. Each record contains: account name (or ID), industry, plan type, MRR, and 2 to 4 behavioral signals (e.g., 'new account, 6 days old', 'usage down 100%'). This list is sent via HTTPS POST to the Anthropic Messages API. Anthropic returns a text response within 30 seconds. The account list is not retained by Anthropic beyond the request per the Business API Data Processing Agreement.

Necessity and proportionality

The LLM explanation step is necessary: without it, the CS team receives a ranked list with raw data signals that require interpretation. The CS team's effectiveness depends on readable, actionable summaries. The processing is proportionate: only aggregated account-level data is sent. No individual user names, email addresses, session identifiers, or personal contact information appear in the prompt.

Risks to data subjects

- Low risk: the data sent is aggregated behavioral data (usage counts, tenure, plan type), not individual personal data.
- Residual risk: if a Ravenstack customer is a sole trader whose company name is their personal name, that name could be sent to Anthropic. The mitigation is to use account IDs instead of names in prompts.

Mitigation measures

- Strip individual personal names from account records before sending to the Anthropic API.
- Use account IDs (e.g., A-310452) rather than company names in prompts if the DPA review identifies a concern.
- Confirm Anthropic DPA covers the types of data sent before Phase 3 go-live.
- Log what data is sent per API call for audit purposes.

Residual risk rating

Low. The data sent is aggregated and behavioral. No special categories of personal data (Article 9) are involved. The Anthropic Business API DPA provides adequate safeguards.

5.4 Data subject rights

Right	How the system supports it
Right of access (Art. 15)	Accounts can request what data Ravenstack holds about them. The scoring model's data comes from Ravenstack's own systems (billing, usage logs, support). The response to an access request draws from those same systems.
Right to erasure (Art. 17)	If an account closes and requests erasure, their records are removed from the scoring API's input data at the next model refresh. The exclusion list in the scoring API config handles this.
Right to rectification (Art. 16)	If account data is incorrect, Ravenstack corrects it in the source system. The scoring API picks up the correction at the next weekly run.
Right to portability (Art. 20)	Not directly applicable (scoring results are internal). Usage logs can be exported on request from Ravenstack's product database.
Right to object (Art. 21)	Accounts can object to profiling for retention purposes. An exclusion list in the scoring API config removes them from future scoring runs.

5.5 Third-party data transfers

Service	Data sent	Legal mechanism	Storage location
Anthropic API	Account name/ID, industry, plan, MRR, behavioral signals (20 accounts per run)	Standard Contractual Clauses (Anthropic Business API DPA)	USA (Anthropic servers). No training retention per DPA.

Slack Technologies	Alert text, account names/IDs, risk scores, LLM explanations	Standard Contractual Clauses (Slack's DPA for paid plans)	USA (Slack data centers). Retention subject to Slack's policy.

Before Phase 3 deployment: both DPAs must be reviewed by someone with legal authority at Ravenstack, signed, and stored on file. If either DPA is inadequate, the legal basis for the transfer must be re-assessed.

6. Strategic deployment and commercialisation plan

6.1 Deployment phases

Phase 1: POC (2 – 4 weeks, complete)

- n8n workflow running locally or on n8n Cloud.
- Scoring runs inside the n8n workflow as a Code node, reading the data from URLs.
- Manual testing with historical Ravenstack data (2023-2024 CSV files).
- Deliverable: working demo that posts a formatted weekly digest to Slack.

Phase 2: Pilot (6 – 10 weeks)

- Scoring API deployed to a production server (e.g., AWS EC2 t3.micro, approximately $10/month).
- n8n workflow active on schedule (Monday 8am).
- CS team receives and rates alerts weekly using Slack emoji reactions.
- Go/no-go decision at week 8: does the model precision reach 50%? If yes, proceed to Phase 3. If no, investigate scoring model before restarting.
- Deliverable: 4 weeks of Slack alerts with CS feedback ratings; measured precision score.

Phase 3: Full deployment (8 – 12 weeks)

- If Phase 2 go: build logistic regression or gradient boosting model using pilot-period data.
- Replace the rule-based logic in the scoring Code node with the trained ML model, or move scoring to a hosted service the workflow calls over a URL. Either way the rest of the workflow does not change.
- Add CRM integration (auto-create tasks in HubSpot or Salesforce when accounts are flagged).
- Set up monitoring dashboard in Power BI (churn trends, alert precision, retention lift).
- Deliverable: live ML-powered system with CRM integration, Slack alerts, and Power BI dashboard.

Phase 4: Scale (ongoing, optional)

- Extend to upsell identification: cross-sell scoring for the acquired company's complementary product.
- Add the 3 upsell use cases identified in the acquisition analysis (error-based cross-sell, tier-mismatch seat expansion, prior-upgrader warm outreach).
- Assess whether to productize the system for other SaaS companies (e.g. “PE Cousins” – other companies back by same PE) after 12 months of measured lift data.

6.2 Go-to-market strategy & commercialization

This system is an internal tool for Ravenstack. The commercialisation question is whether to remain internal or productize it for other SaaS companies after proving lift.

As an internal tool (Phases 1-3)

- No external go-to-market is required.
- Cleo approves the pilot budget ($10,000) and the Phase 3 ongoing cost ($1,395/month).
- Success is measured by Ravenstack's own churn rate change, not external revenue.

If productized (Phase 4 option)

- **Target buyers:** CS leaders and RevOps managers at "**PE-Cousins**" (B2B SaaS companies of the same PE or VC) and no internal ML team.
- **Sales channel:** Direct outbound to CS and RevOps leaders at SaaS companies; LinkedIn and SaaS community forums (Pavilion, RevOps Co-op).
- **Pricing model:** SaaS, $500 to $1,000 per month depending on account volume. Annual contracts preferred to reduce the system's own churn.
- **Key differentiator:** The enterprise retention platforms (ChurnZero, Vitally, Pecan, Gainsight, Totango) start at $50,000 to $100,000 per year. This product targets companies getting churn & upsell flagging quickly without the overhead of a full platform implementation.

The recommendation for now is to complete Phases 1 to 3 as an internal tool, measure actual lift over 12 months, and revisit productization with real performance data. Building a product before proving the lift would be premature.

6.3 Stakeholder communication plan

Stakeholder	What they need to know	Who communicates	When
Cleo (CEO)	Pilot budget approved, timeline, what success looks like, cost at each phase gate	Markus	Before Phase 2; monthly status updates; go/no-go at week 8
CS team lead	Why the digest exists, how to rate alerts, time commitment (5 minutes/week), how to flag errors	Markus + CS lead briefing	Before first live alert; weekly check-in for 4 weeks; then monthly
CS team members	How to read the Slack digest, what the risk score means, that they decide all contact	CS team lead	Before first live alert; 15-minute training session

Engineering	How to maintain the scoring API, what to do if it breaks, monthly retraining process, drift detection alerts	Markus via runbook documentation	Before Phase 3 goes live

6.4 KPIs per phase

Phase 1 (POC)

- Workflow runs without error on manual test execution.
- Slack digest arrives with correct formatting and 20 account records visible.
- Each record shows a score, a visual bar, and a plain-English explanation.

Phase 2 (Pilot)

- Precision: CS team rates ≥50% of flagged accounts as relevant by week 4.
- Reaction rate: ≥80% of alerts receive a Slack reaction (✅ or ⏭️) within 24 hours of posting.
- Model precision vs control: at week 8, compare churn rate of flagged accounts that received outreach versus those that did not. Target: 15% lower churn in the outreach group.

Phase 3 (Full deployment)

- Monthly churn rate reported in Power BI dashboard: track whether the 12.1% rate falls.
- ARR retained from flagged accounts: target $7,500/month (to match the ROI model).
- CRM task completion rate: ≥70% of auto-created tasks resolved within 7 days.

Phase 4 (Scale)

- Total ARR retained versus the pre-system 12-month baseline.
- If productized: customer acquisition cost per new SaaS customer, monthly recurring revenue, and net revenue retention.